

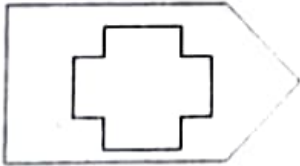


Academic year 2025-2026 (ODD Sem)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Date: Jan/2026	CIE – 1 Retest	Max. Marks:(10+50) = 60
Semester: VII	UG	Duration: 2 Hours
Course Title: Computer Graphics & Virtual Reality		Course Code: CS373IA

Q. No.	QUIZ	M	BT	CO
1.	What are the stages of the Graphics Pipeline in OpenGL?	02	2	2
2.	What are Primitives in OpenGL? Give examples.	02	2	1
3.	Define a State Machine in the context of OpenGL.	01	3	1
4.	What are the Three I's of Virtual Reality?	01	2	1
5.	What are the five classic components of a VR system?	01	2	2
6.	Define Display List and mention its advantage.	02	3	2
7.	Differentiate between RGB color and Indexed color.	01	2	1

Q. No.	TEST	M	BT	CO
1.	GLUT implements event-driven programming model for user window. For example, when the user clicks or moves the mouse, GLUT can call our <i>event handler function</i> and allow the user's actions. Now identify GLUT callback functions to perform the following operations: when window needs to be drawn, when there is nothing else to do, when the key goes up when mouse drags around, when mouse moves Explain the parameters used and the purpose of each of the above calls	10	3	2
2.	Write a program to recursively subdivides a tetrahedron to form 3D Sierpinski gasket. The number of recursive steps is to be specified at execution time.	10	3	2
3a.	Write an OpenGL complete program to display the below figure on the window, identify the proper coordinates in drawing the figure using only Line Loop primitive with inner figure being in RED color and outer GREEN in color. 	5	3	2
3b.	Any application program can obtain the measure of a device in three distinct modes, with the help of relevant examples and diagrams explain three input modes.	5	2	1



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4.	Write a program to demonstrate the approximation of a 3D sphere with appropriate mathematical formulations. Write the complete C program to approximate a sphere using OpenGL primitives.	10	3
5.	Explain the working of a 3D Positional Tracker and discuss the major Tracker Performance Parameters that affect its accuracy and usability.	10	3

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Apply the basic concepts of Computer Graphics & Virtual Reality which illustrates the pipeline architecture, OpenGL library, VR tools.
CO2:	Analyze and make an appropriate choice of methods required for computer representation of objects and modelling in Virtual Reality.
CO3:	Design Computer Graphics and Virtual Reality applications using OpenGL library & VR tools.
CO4:	Implement common geometric construction & VR techniques and develop Computer Graphics based solution to Engineering applications / real-world problems.
CO5:	Exhibit team work and effective oral/written communication skills to accomplish a community project of solving Computer Graphics/Virtual Reality problems with the engineering community and society at large.